

Full-stack software engineer with 4+ years of experience building backend systems, operational tooling, and distributed services.

PROFESSIONAL EXPERIENCE

Software Engineer, Persona (2022 - 2026)

Identity Data Products

- Architected and implemented a custom Elasticsearch retrieval system that replaced $O(n^2)$ CPU-intensive processing workflows, reducing execution times from hours to minutes or seconds while improving scalability
- Led planning and execution of data migrations involving up to 19 million accounts, designing rollout, monitoring, validation, and rollback strategies to safely migrate a customer representing up to 50% of production traffic.
- Delivered compliance and identity data product enhancements in collaboration with product, solutions engineering, and operational stakeholders.

Cases

- Redesigned case routing architecture by separating queueing and assignment policies into independent rule systems, enabling configurable multi-stage review workflows and improved workload balancing across investigation teams.
- Designed and implemented SLA tracking infrastructure supporting flexible deadline configuration and business-hour aware expiration calculations for investigation workflows.
- Built a reusable notification framework supporting multi-channel delivery, user-configurable preferences, and scheduled notification workflows, enabling operational features such as SLA expiration alerts.
- Built customer outreach and document collection capabilities for manual case review, centralizing communications and supporting documentation within case records

Identity Labs

- Supported identity verification and risk assessment initiatives through development of scoring models and supporting infrastructure.

Software Engineer, Bloomberg (2021 - 2022)

News Content Team

- Developed newsroom tooling and content experiences supporting coverage of Bloomberg's New Economy Forum.
- Redesigned news article export workflows to process asynchronously, improving responsiveness of print and PDF generation.
- Improved frontend performance by reducing unnecessary Redux subscriber updates and state-change propagation.

EDUCATION

University of Waterloo (2016 - 2021)

Bachelors of Mathematics, Honours Computer Science

SKILLS

Languages

- Ruby
- Typescript
- SQL
- Python

Frameworks & Technologies

- Ruby on Rails
- React
- Redux
- Elasticsearch
- MySQL
- Redis
- Sidekiq
- Datadog
- Docker
- Git
- REST APIs
- RSpec